

KUNGRAD, UzbekistanWMO: **381490**Lat: **43.083N**Lon: **58.933E**Elev: **197**StdP: **14.59**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	1.6	-10.8	3.1	-3.1	-5.2	4.2	2.2	32.8	26.9	26.4	29.9	5.0	10	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	102.8	70.7	99.8	69.7	97.1	69.0	72.8	96.5	71.5	95.6	70.2	93.8	8.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	92.8	83.7	62.9	86.5	82.8	61.3	81.6	82.1	36.5	97.0	35.3	95.8	34.2	94.1	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.1	23.2	18.9	-7.6	107.4	8.0	3.1	-13.3	109.7	-18.0	111.5	-22.4	113.2	-28.2	115.4
			-8.1	76.0	7.8	1.7	-13.7	77.2	-18.3	78.3	-22.6	79.2	-28.2	80.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	24.1	28.0	42.3	57.7	70.8	79.3	82.5	78.9	66.6	52.6	37.9	27.7	(6)
(7)		DBStd	22.46	10.69	10.87	10.01	9.58	6.95	5.85	5.14	6.11	7.02	8.62	9.69	9.16	(7)
(8)		HDD50	2874	804	618	277	36	0	0	0	0	1	73	374	691	(8)
(9)		HDD65	5719	1269	1037	703	256	28	2	0	1	63	392	812	1156	(9)
(10)		CDD50	4399	0	1	40	268	646	879	1007	894	499	153	12	0	(10)
(11)		CDD65	1766	0	0	1	37	208	430	542	430	111	7	0	0	(11)
(12)		CDH74	24071	0	1	25	506	2599	5938	7630	5783	1449	139	1	0	(12)
(13)		CDH80	13245	0	0	5	171	1144	3409	4585	3358	552	21	0	0	(13)
(14)	Wind	WSAvg	8.0	7.4	8.4	9.7	9.8	9.3	8.5	7.4	7.1	7.0	7.0	7.3	(14)	
(15)	Precipitation	PrecAvg	4.5	0.4	0.4	0.6	0.7	0.6	0.3	0.1	0.1	0.2	0.2	0.5	0.4	(15)
(16)		PrecMax	9.4	1.0	1.9	1.9	2.1	3.0	1.9	0.5	0.5	0.7	1.3	2.2	1.2	(16)
(17)		PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.9	0.2	0.4	0.5	0.5	0.7	0.4	0.2	0.1	0.2	0.3	0.6	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	59.8	77.1	90.7	97.8	105.2	107.8	96.6	82.9	68.3	53.7	(19)	
(20)			MCWB	43.3	49.7	58.6	65.3	68.1	71.2	72.1	71.9	68.8	62.0	56.0	45.7	(20)
(21)		2%	DB	43.8	51.7	69.2	85.1	93.5	101.2	103.3	101.6	90.0	78.1	61.5	47.1	(21)
(22)			MCWB	38.6	44.4	55.5	63.6	67.1	70.0	70.8	70.3	66.4	59.5	50.8	40.9	(22)
(23)		5%	DB	40.0	46.2	63.6	79.8	90.1	98.0	100.2	98.2	86.1	74.0	56.9	43.0	(23)
(24)			MCWB	35.8	40.1	52.2	61.1	65.8	69.1	69.7	69.6	64.5	58.0	48.0	38.4	(24)
(25)		10%	DB	36.8	41.6	58.5	74.8	86.6	95.0	97.3	94.9	82.5	69.1	52.3	39.7	(25)
(26)			MCWB	33.3	36.6	49.1	58.7	64.7	68.3	69.1	68.4	63.1	55.2	45.4	35.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.8	50.6	60.1	67.7	71.5	74.6	74.7	74.6	70.0	63.6	57.0	46.0	(27)
(28)			MCD B	49.2	59.3	76.0	85.1	91.5	96.2	98.3	97.9	91.1	80.2	67.1	52.2	(28)
(29)		2%	WB	39.1	44.4	55.6	64.6	69.2	72.1	72.7	72.2	67.4	60.8	51.9	41.6	(29)
(30)			MCD B	43.1	51.5	68.4	82.5	89.2	95.8	96.9	96.6	88.2	75.9	60.2	46.5	(30)
(31)		5%	WB	36.1	40.3	52.5	62.0	67.4	70.7	71.4	70.7	65.4	58.2	48.6	38.5	(31)
(32)			MCD B	39.7	45.9	63.3	78.6	86.8	94.5	96.0	95.0	84.5	72.7	55.8	42.8	(32)
(33)		10%	WB	33.6	36.8	49.3	59.3	65.5	69.2	70.2	69.3	63.4	55.8	45.8	35.7	(33)
(34)			MCD B	36.6	41.3	58.1	73.5	84.3	92.0	94.0	92.8	81.2	68.2	52.3	39.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.9	18.0	21.1	22.8	24.4	26.3	26.4	28.3	27.4	25.6	19.3	15.2	(35)
(36)			MCDBR	17.2	24.5	28.8	28.8	29.2	30.7	30.4	32.9	31.7	31.6	26.7	20.3	(36)
(37)			MCWBR	13.5	18.7	18.2	14.5	12.3	10.9	9.6	11.8	14.6	17.6	18.2	16.0	(37)
(38)		5% WB	MCDBR	16.0	23.4	28.0	27.5	27.1	28.0	27.7	30.9	31.5	29.3	24.4	19.1	(38)
(39)			MCWBR	12.8	18.1	18.1	15.4	13.7	11.8	10.7	12.5	15.5	16.8	17.3	15.2	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.379	0.405	0.470	0.525	0.529	0.513	0.497	0.472	0.445	0.450	0.411	0.382	(40)
(41)		taud		2.072	2.019	1.909	1.814	1.851	1.920	1.988	2.026	2.039	1.971	2.036	2.064	(41)
(42)		Ebn at Noon		224	240	239	237	241	246	248	249	245	221	208	208	(42)
(43)		Edh at Noon		34	43	54	64	64	60	55	51	47	44	36	32	(43)
(44)	All-Sky Solar Radiation	RadAvg		476	797	1243	1708	2103	2358	2272	2082	1643	1083	641	432	(44)
(45)		RadStd		55	96	101	118	116	103	84	75	68	63	77	58	(45)